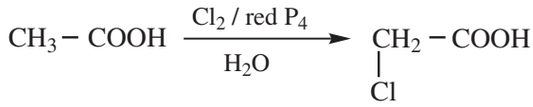
D) Substitution reactions in the hydrocarbon part

1) α - Halogenation

Carboxylic acids having an α - hydrogen are halogenated at the α - position on treatment with chlorine or bromine in the presence of small amount of red posphorus to form α halo carboxylic acids. This reaction is known as **Hell – Volhard – Zelinsky reaction** (HVZ reaction) The α - Halogenated acids are convenient starting materials for preparing α - substituted acids.

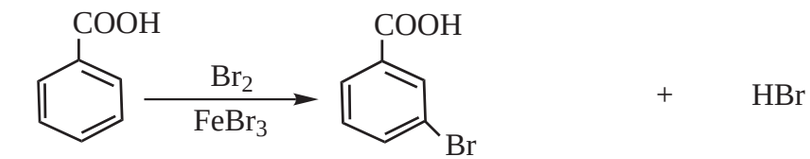


Acetic acid — Mono Chloro acetic acid

2) Electrophilic substitution in aromatic carboxylic acids

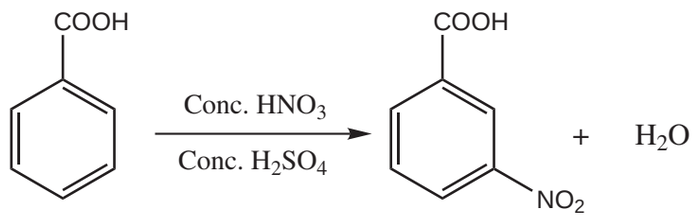
Aromatic carboxylic acid undergoes electrophilic substitution reactions. The carboxyl group is a deactivating and meta directing group. Some common electrophilic substitution reactions of benzoic acid are given below

i) Halogenation



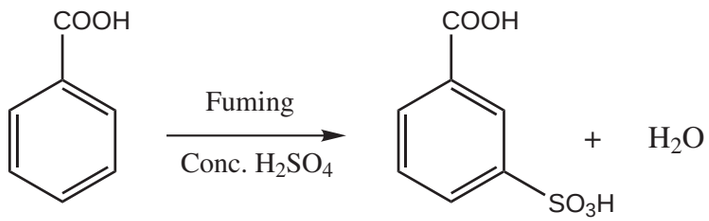
Benzoic acid — m - bromo benzoic acid

ii) Nitration



Benzoic acid — m-Nitrobenzoic acid

iii) Sulphonation



Benzoic acid — m-Sulphobenzoic acid

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